

5-Mark Questions & Answers -- DBMS Module 3: SQL Standards & Advanced Queries

Q1. Explain Nested Subqueries (IN, NOT IN, EXISTS, NOT EXISTS, ALL, ANY) with SQL examples.

* IN / NOT IN: Checks membership in subquery result set.

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Q2. Explain Recursive Queries and Common Table Expressions (CTE) in SQL.

* CTE Syntax: WITH cte_name AS (query) SELECT * FROM cte_name;

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WITH RECURSIVE EmpHierarchy AS (  
  -- Anchor Member (Top Manager)  
  SELECT Emp_ID, Name, Manager_ID, 1 AS Level FROM Employee WHERE Manager_ID IS NULL  
  UNION ALL  
  -- Recursive Member  
  SELECT E.Emp_ID, E.Name, E.Manager_ID, H.Level + 1  
  FROM Employee E INNER JOIN EmpHierarchy H ON E.Manager_ID = H.Emp_ID  
)  
SELECT * FROM EmpHierarchy;
```

Q3. Explain PL/SQL Triggers: Syntax, Timing, Events, and Use Cases.

* Complete syntax for CREATE OR REPLACE TRIGGER with BEFORE/AFTER, FOR EACH ROW, and :NEW / :OLD qualifiers for audit logging and data validation.